

## STAGE 2 CHEMISTRY · FULL PRACTICE EXAM

# Chemistry Practice Exam

Written by TopMark Tutors, mapped to the SACE Stage 2 Chemistry subject outline

TOTAL MARKS

**120**

TOTAL TIME

**130 min**

BOOKLET 1

**Q1–3 · 63 marks**

BOOKLET 2

**Q4–7 · 57 marks****Topics covered**

- Topic 1 — Monitoring the Environment
- Topic 2 — Managing Chemical Processes
- Topic 3 — Organic and Biological Chemistry
- Topic 4 — Managing Resources

**Instructions**

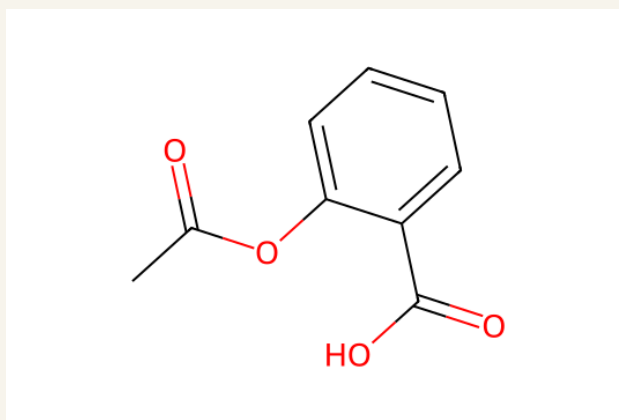
- Questions 1 to 3 (63 marks) — answer all questions in this booklet, allow approximately 65 minutes
- Show all working for calculations — method marks are available
- A periodic table and standard data sheet may be used
- Approved calculators may be used

Name: \_\_\_\_\_

This is an original practice paper written by TopMark Tutors for study purposes. It is built to reflect the style, structure, and topic weighting of real SACE Stage 2 Chemistry examinations, but is not a past SACE exam paper and is not produced, endorsed, or reviewed by the SACE Board of South Australia. All questions, data, and scenarios are original. A full marking guide is available — see the final page.

## TOPIC 3 · ORGANIC &amp; BIOLOGICAL CHEMISTRY

**Question 1** (21 marks) — Aspirin (acetylsalicylic acid) is synthesised from salicylic acid and is widely used as a pain reliever. Its structural formula is shown below.

Aspirin,  $C_9H_8O_4$ 

(a) Identify the two functional groups present in aspirin.

(2 marks)

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(b) Aspirin is industrially synthesised by reacting salicylic acid with ethanoic anhydride, rather than with ethanoic acid directly.

(i) Draw the structural formula of ethanoic anhydride,  $(CH_3CO)_2O$ .

(1 mark)

(ii) Suggest one advantage of using ethanoic anhydride rather than ethanoic acid for this synthesis.

(1 mark)

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*Question 1 continued*

(c) Aspirin can be hydrolysed under acidic conditions to reform salicylic acid and a second product.

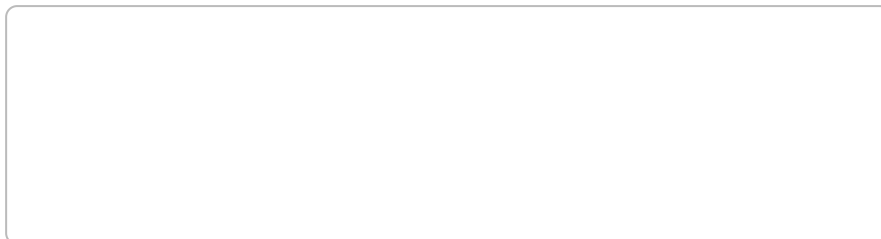
(i) Name the type of reaction occurring when aspirin is hydrolysed.

**(1 mark)**

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(ii) Draw the structural formula of the second product formed during this hydrolysis.

**(2 marks)**



Question 1 continued

(d) Salicylic acid is more soluble in water than aspirin. Explain this difference in solubility with reference to the functional groups present in each molecule.

(3 marks)

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(e) State one chemical test, and the expected observation, that would distinguish salicylic acid from aspirin.

(2 marks)

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(f) HPLC was used to analyse a tablet containing both aspirin and caffeine. Caffeine is more polar than aspirin, and a non-polar stationary phase was used in the column. Explain which compound would elute from the column first, with reference to the principles of chromatography.

(3 marks)

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(g) Aspirin has a molar mass of  $180.16 \text{ g mol}^{-1}$ . Calculate the percentage by mass of oxygen in aspirin,  $\text{C}_9\text{H}_8\text{O}_4$ .

(3 marks)

(h) Glycine and alanine are amino acids that can combine to form a dipeptide.

(i) Draw the structural formula of the dipeptide formed when glycine and alanine react, clearly showing the peptide (amide) bond.

(2 marks)

(ii) State the number of water molecules released when three amino acid molecules combine to form a tripeptide.

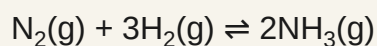
(1 mark)

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Total for Question 1: 21 marks

## TOPIC 2 · MANAGING CHEMICAL PROCESSES

**Question 2** (20 marks) — Ammonia is produced industrially by the Haber process, shown below.

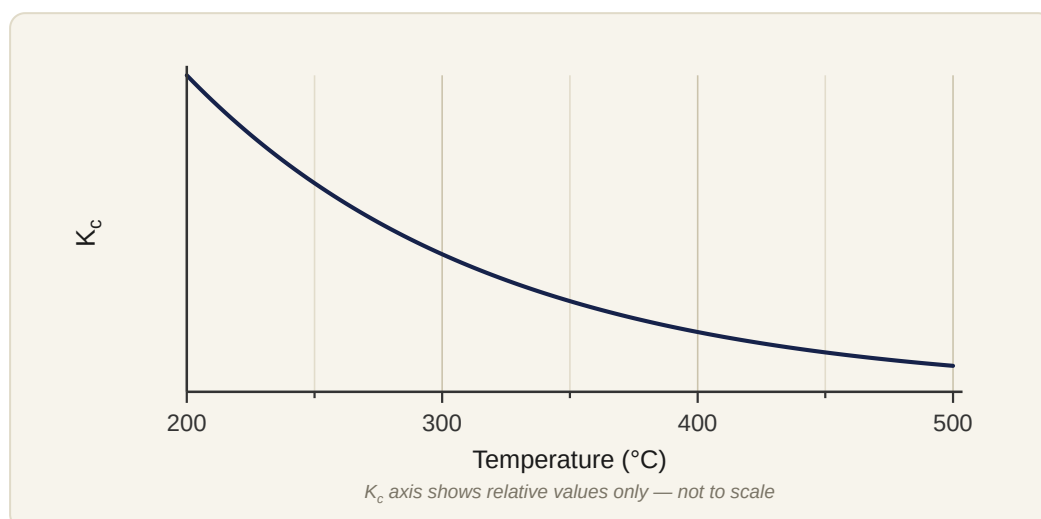


(a) A 2.00 L reaction vessel initially contained 1.20 mol of  $\text{N}_2$  and 3.60 mol of  $\text{H}_2$ . At equilibrium, 0.80 mol of  $\text{NH}_3$  was present. Calculate the value of  $K_c$  for this reaction at this temperature. Show all working.

(5 marks)

(b) The graph below shows how  $K_c$  for the Haber process changes with temperature.

(3 marks)



Using the graph, state and explain whether the forward reaction of the Haber process is exothermic or endothermic.

Question 2 continued

(c) Industrially, the Haber process is run at approximately 450°C rather than at a lower temperature, even though a lower temperature would favour a higher equilibrium yield of ammonia. Explain one reason for this choice.

(3 marks)

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(d) State and explain the effect of increasing the pressure on the equilibrium position of the Haber process.

(2 marks)

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Question 2 continued

(e) An iron catalyst is used in the Haber process. Explain the effect of this catalyst on the value of  $K_c$  and on the rate at which equilibrium is reached.

(2 marks)

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(f) Once equilibrium has been established, additional  $N_2(g)$  is injected into the reaction vessel at constant volume and temperature. State the direction in which the equilibrium will shift, and explain your reasoning.

(2 marks)

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(g) If hydrogen gas is continuously removed from the equilibrium mixture as it forms, state and explain the effect this would have on the yield of ammonia over time.

(3 marks)

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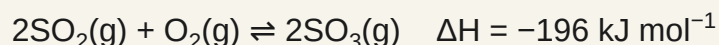
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## TOPIC 2 · MANAGING CHEMICAL PROCESSES

**Question 3** (22 marks) — Sulfuric acid is manufactured using the Contact Process. A key step is the catalytic oxidation of sulfur dioxide:



- (a) A vanadium(V) oxide catalyst is used for this reaction. Explain, in terms of activation energy and reaction mechanism, how the catalyst increases the rate of this reaction. (3 marks)

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- (b) The Contact Process is operated at close to atmospheric pressure, even though the forward reaction converts 3 mol of gas into 2 mol. Suggest one reason for this choice. (1 mark)

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- (c) Sulfur dioxide released from industrial processes contributes to acid rain.

- (i) Write an equation showing how  $\text{SO}_2$  reacts with water in the atmosphere to form an acidic solution. (2 marks)

- (ii) Write an equation showing how the acid formed in part (i) reacts with limestone ( $\text{CaCO}_3$ ) in soil and buildings. (2 marks)



Question 3 continued

(d) Explain, using an equilibrium argument, why increased acidity in freshwater lakes is harmful to shell-forming aquatic organisms.

(3 marks)

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(e) A factory releases waste gas containing  $\text{SO}_2$  at a concentration of 340 ppm. Calculate this concentration in  $\text{mg L}^{-1}$ , assuming the density of the waste gas is equivalent to  $1.20 \text{ g L}^{-1}$ .

(2 marks)

(f) Explain one economic benefit to the manufacturer of capturing and reacting excess  $\text{SO}_2$  rather than releasing it as waste.

(2 marks)

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(g) Modern Contact Process plants use a “double absorption” step, where  $\text{SO}_3$  is removed and re-absorbed partway through the process rather than in a single pass. With reference to one or more relevant Science as a Human Endeavour concepts, discuss one advantage and one limitation of using double absorption in the Contact Process.

(4 marks)

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(h) State one commercial use of sulfuric acid produced by the Contact Process.

(1 mark)

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(i)  $\text{NO}_x$  emissions from industrial combustion also contribute to acid rain. Write one equation showing the formation of NO from atmospheric  $\text{N}_2$  and  $\text{O}_2$  at high temperature.

(2 marks)

## STAGE 2 CHEMISTRY · FULL PRACTICE EXAM

# Chemistry Practice Exam

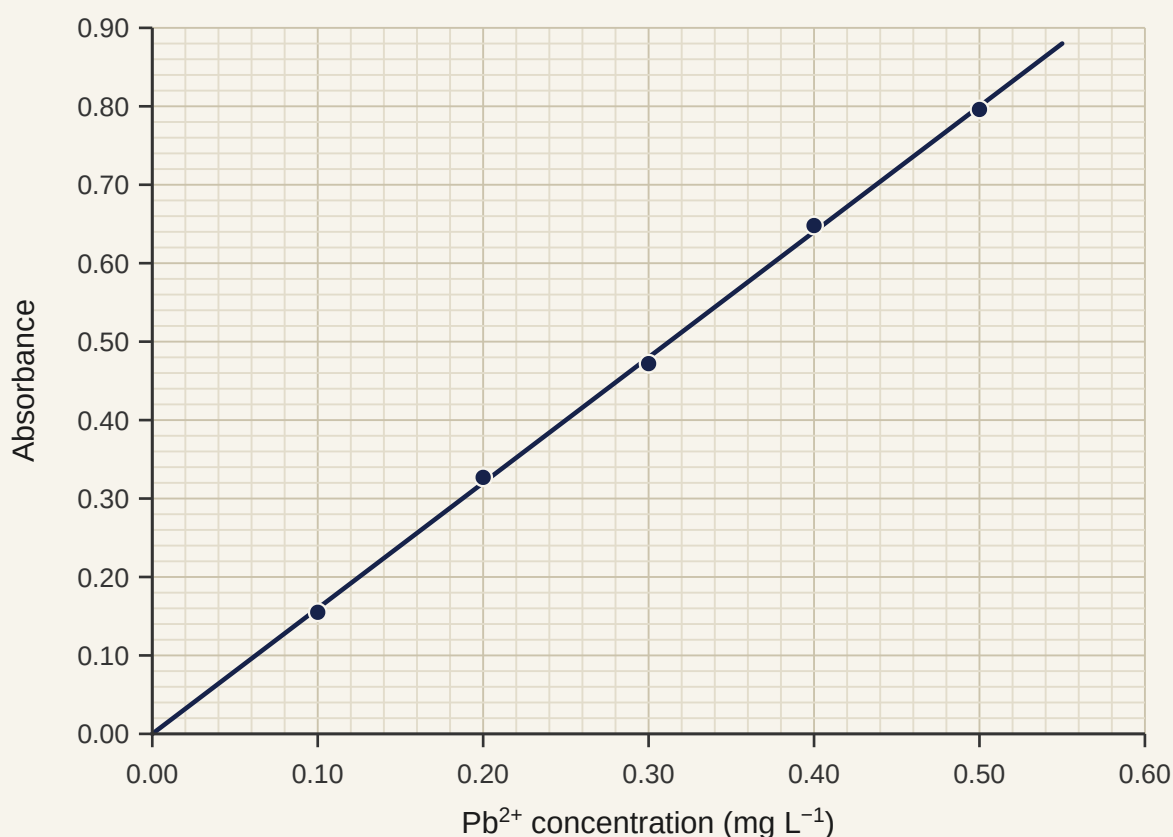
Written by TopMark Tutors, mapped to the SACE Stage 2 Chemistry subject outline

**This booklet**

- Questions 4 to 7 (57 marks) — answer all questions in this booklet, allow approximately 65 minutes
- Show all working for calculations — method marks are available
- A periodic table and standard data sheet may be used
- Approved calculators may be used

## TOPIC 1 · MONITORING THE ENVIRONMENT

**Question 4** (15 marks) — Lead contamination in groundwater drawn for drinking supply is monitored using Atomic Absorption Spectroscopy (AAS). Five standard solutions of lead nitrate were used to construct the calibration graph below.



(a) A groundwater sample produced an absorbance reading of 0.62. Using the calibration graph, estimate the lead concentration, in mg L<sup>-1</sup>, in this sample.

(1 mark)

(b) Explain, in terms of electron transitions, why atoms of lead absorb only specific wavelengths of light during AAS analysis.

(2 marks)

Question 4 continued

(c) The Australian drinking water guideline for lead is a maximum of  $10 \mu\text{g L}^{-1}$ . Convert your answer from part (a) into  $\mu\text{g L}^{-1}$ , and state whether this sample exceeds the guideline.

(2 marks)

(d) State the term used to describe a measured result that is close to the true concentration of the sample.

(1 mark)

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(e) Other dissolved ions in the groundwater sample can interfere with AAS readings, causing a systematic error. Describe how the technique of standard addition can be used to correct for this type of interference.

(3 marks)

(f) A separate water sample was tested for dissolved chloride ions using a titration with standardised silver nitrate solution. A  $25.00 \text{ mL}$  sample required an average titre of  $14.20 \text{ mL}$  of  $0.0500 \text{ mol L}^{-1} \text{ AgNO}_3$ .

(i) Write the ionic equation for the reaction occurring in this titration.

(1 mark)

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(ii) Calculate the concentration of chloride ions, in  $\text{mol L}^{-1}$ , in the original sample.

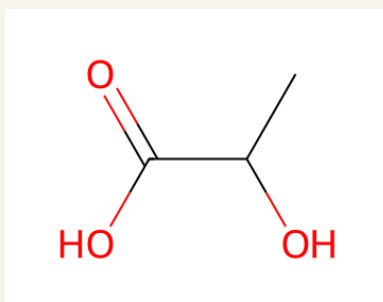
(3 marks)

(iii) Hence, calculate the concentration of chloride ions in this sample in  $\text{mg L}^{-1}$ .

(2 marks)

## TOPIC 3 · ORGANIC &amp; BIOLOGICAL CHEMISTRY

**Question 5** (14 marks) — Polylactic acid (PLA) is a biodegradable polymer made from lactic acid monomers and is increasingly used to replace conventional plastics in packaging.



Lactic acid monomer

(a) Identify the two functional groups present in the lactic acid monomer.

(2 marks)

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(b) Identify the type of polymerisation that produces PLA from lactic acid monomers, and name the small molecule released as a by-product of each linkage formed.

(2 marks)

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(c) Draw the structural formula of a section of PLA containing two lactic acid monomer units, joined by one ester linkage.

(2 marks)

(d) PLA can be broken down by hydrolysis under composting conditions, unlike conventional polyethylene. Explain, with reference to the structure of each polymer, why PLA is biodegradable while polyethylene is not.

(3 marks)

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Question 5 continued

(e) One sample of PLA has an average degree of polymerisation of 1200. Given that the molar mass of one lactic acid monomer unit is  $90.08 \text{ g mol}^{-1}$ , and each condensation reaction releases one water molecule ( $18.02 \text{ g mol}^{-1}$ ), estimate the average molar mass of this PLA polymer.

(3 marks)

(f) With reference to one relevant Science as a Human Endeavour concept, discuss one limitation of using PLA as a replacement for conventional plastic packaging.

(2 marks)

TOPIC 4 · MANAGING RESOURCES

**Question 6** (14 marks) — Copper is refined industrially using electrolysis, and its concentration in solution can be verified by iodometric titration.

(a) During the electrolytic refining of copper, an impure copper anode dissolves and pure copper is deposited at the cathode.

(i) Write the half-equation for the reaction occurring at the cathode. (1 mark)

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(ii) Explain why this process is described as electrolytic refining rather than electrolytic production. (2 marks)

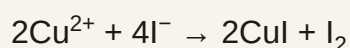
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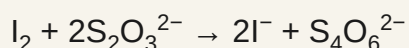
(iii) Using subshell notation, write the electron configuration of the  $\text{Cu}^{2+}$  ion. (2 marks)

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(b) A 20.00 mL sample of copper(II) sulfate solution was acidified and treated with excess potassium iodide, producing iodine according to the equation:



The iodine produced was titrated with  $0.1000 \text{ mol L}^{-1}$  sodium thiosulfate solution, requiring an average titre of 18.40 mL, according to the equation:



(i) Calculate the number of moles of thiosulfate ions used in the titration. (1 mark)

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Question 6 continued

(ii) Hence calculate the concentration of  $\text{Cu}^{2+}$ , in  $\text{mol L}^{-1}$ , in the original 20.00 mL sample. Express your answer to an appropriate number of significant figures.

(4 marks)

(c) The impure copper anode also contains small amounts of silver and gold, which do not dissolve during electrolysis and instead form a solid residue called “anode sludge”. Explain why this anode sludge has economic value to the refinery, and identify why silver and gold do not dissolve at the anode.

(3 marks)

(d) State one commercial use of the hydrogen gas produced at the cathode during the electrolysis of brine (the chlor-alkali process).

(1 mark)



## TOPIC 1 · MONITORING THE ENVIRONMENT

**Question 7** (14 marks) — Excess nitrate from agricultural fertiliser can leach into waterways, contributing to eutrophication of lakes and rivers.

- (a) Explain, in terms of algal growth and dissolved oxygen levels, how excess nitrate entering a lake leads to eutrophication and can result in the death of fish populations. (3 marks)

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- (b) One method of removing nitrate from contaminated water is to pass it through an ion exchange column. State the type of charge on the resin used to remove nitrate anions. (1 mark)

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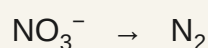
- (c) A water sample was found to contain nitrate at a concentration of 62 ppm. Calculate this concentration in  $\text{mol L}^{-1}$ , assuming the density of the sample is  $1.00 \text{ g mL}^{-1}$ . (2 marks)

- (d) Unlike phosphate, nitrate cannot easily be removed from water by precipitation with calcium or aluminium salts. Suggest one reason for this, with reference to the solubility of common nitrate compounds. (2 marks)

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- (e) Denitrifying bacteria in soil can convert nitrate ions into nitrogen gas. Complete and balance the half-equation below for this reduction reaction in acidic conditions. (3 marks)



Please turn over →

Question 7 continued

(f) With reference to one relevant Science as a Human Endeavour concept, discuss one social or economic implication for farmers of introducing stricter regulations on fertiliser use to protect waterways.

**(3 marks)**

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Total for Question 7: 14 marks

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**End of Question Booklet 2 • Total: 57 marks**

Combined total for this exam: 120 marks

## MARKING GUIDE

# A separate marking guide is available with this paper.

### The SACE Exam Crash Course

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